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Where in the oscillating motion of a mass on a spring, is the net force the greatest?

- ☐ Half way between the equilibrium position and greatest extension.
- ☐ At the greatest extension.
- ☐ At the equilibrium position (middle).
- ☐ At the greatest compression.
- ☐ At the greatest compression and greatest extension.

When does the kinetic energy (KE) equal the gravitational potential energy (GPE) during a simple pendulum's oscillation?

- ☐ At the bottom of the pendulum's motion.
- ☐ Half-way between its maximum height and its lowest point.
- ☐ Never.
- ☐ At the top of its motion.

During the simple harmonic motion of a pendulum, where is the acceleration greatest?

- ☐ At the far right only.
- ☐ At the equilibrium position (middle).
- ☐ At the far left only.
- ☐ At the far left and far right.
- ☐ It is the same throughout the motion.

Create the imagen in your brain that you are driving your car (picture a nice one ;)) on a parkway which has bumps every ***30 m*** apart. At what speed must you be driving to experience violent shaking if the suspension in your car has a natural frequency of ***0.5 Hz?***

Having three identical springs connecting two identical masses in between. Combine the equation of motion of the two masses to obtain a pair of equations of the form:

$$\frac{d^2q_1}{dt^2} + \omega_1^2 q_1 = 0 \text{ and } \frac{d^2q_2}{dt^2} + \omega_2^2 q_2 = 0$$



and hence obtain the normal coordinates q_1 and q_2 and the respective normal frequencies ω_1 and ω_2

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Having a coupled oscillator consisting of two equal masses connected by three identical springs of length ***l*** and under tension ***T*** (consider tension as a constant), as shown in Figure. The masses are displaced in the transverse direction by distances of y_a and y_b, respectively. Find the normal frequencies in terms of tension, mass and spring length. Consider a Euler equation for the transverse displacements.

